

FOOD RESOURCES

The main sources of human food are plants and animals. We consume almost all parts of one or the other plant in the form of cereals, pulses, vegetables, fruits and spices. A number of animal products such as milk, butter, egg and meat also supplement our food requirements.

World Food Problems – World's population is growing every year and so the demand of food is increasing constantly. Although world's food production has increased almost three times during the last 50 years, but at the same time rapid population growth especially in less developed countries has outstripped the food production. About 40 million people die every year due to undernourishment and malnutrition.

The major source of food are crops. The cropping pattern and crop yield are determined mainly by natural factors like rainfall, climate and soil conditions.

Impacts of Overgrazing

1. Overgrazing removes the vegetal cover of the soil and makes it compact. It reduces the humus content due to decline in organic cycling and infiltration capacity of the soil. As a result more water is lost from the ecosystem along with surface runoff and the soil loses its fertility.
2. Due to the loss of vegetal cover the soil becomes susceptible to the action of wind and water and gets eroded.
3. Overgrazing affects the composition of plant population and their generation capacity.

Impacts of Traditional Agriculture – About half of global population practice traditional agriculture. It involves small fields simple implements, naturally available water, organic fertilizers and mix crops and results in low production.

1. Shifting cultivation (Slash and burn cultivation) practiced in many tribal areas results in deforestation.
2. Loss of forest cover exposes the soil to wind, rain and storms, thereby resulting in soil erosion.
3. Slash and burn cultivation destroys the organic matter and makes the soil nutrient poor within a short period.

Impacts of Modern Agriculture – Modern agriculture is largely based upon technological factors like the use of improved seeds, chemical fertilizers, synthetic pesticides and extensive irrigation.

1. Intensive farming have reduced the fertility and productivity of the soil without the use of chemical fertilizers.
2. Fertilizers have contaminated groundwater with nitrate. Nitrate reacts with haemoglobin and impairs the oxygen transport by the blood for infants. This condition is called **methaemoglobinemia** or **blue baby syndrome**.
3. Excessive NPK fertilizers used in agriculture fields are often washed off with runoff water to water bodies and lakes causing overnourishment of the lakes called **Eutrophication**. **Eutrophication** often leads to algal blooms, which make water unfit for consumption and kill the aquatic life.
4. Some individuals of the pest species survive even after pesticide spray and give rise to highly resistant super pests which are immune to all types of pesticides.

5. Intensive irrigation is bringing the underground soluble salts to the soil surface and increasing the salinity of the soil. A large area of fertile land have become saline and water logged in the recent years due to excessive irrigation.